MILESTONE 4 – COVER PAGE

Team Number: 22

Please list full names and MacID's of all *present* Team Members

Full Name:	MacID:
Matthew Ho	Hom23
Ruidi Liu	liu127
Mayar Aljayoush	aljayoum
Sharanya Srirangam	srirans

MILESTONE 4 (STAGE 1) – COMPUTER PROGRAM WORKFLOW (COMPUTATION SUB-TEAM)

Team Number: 22

You should have already completed this task individually *prior* to Design Studio 17.

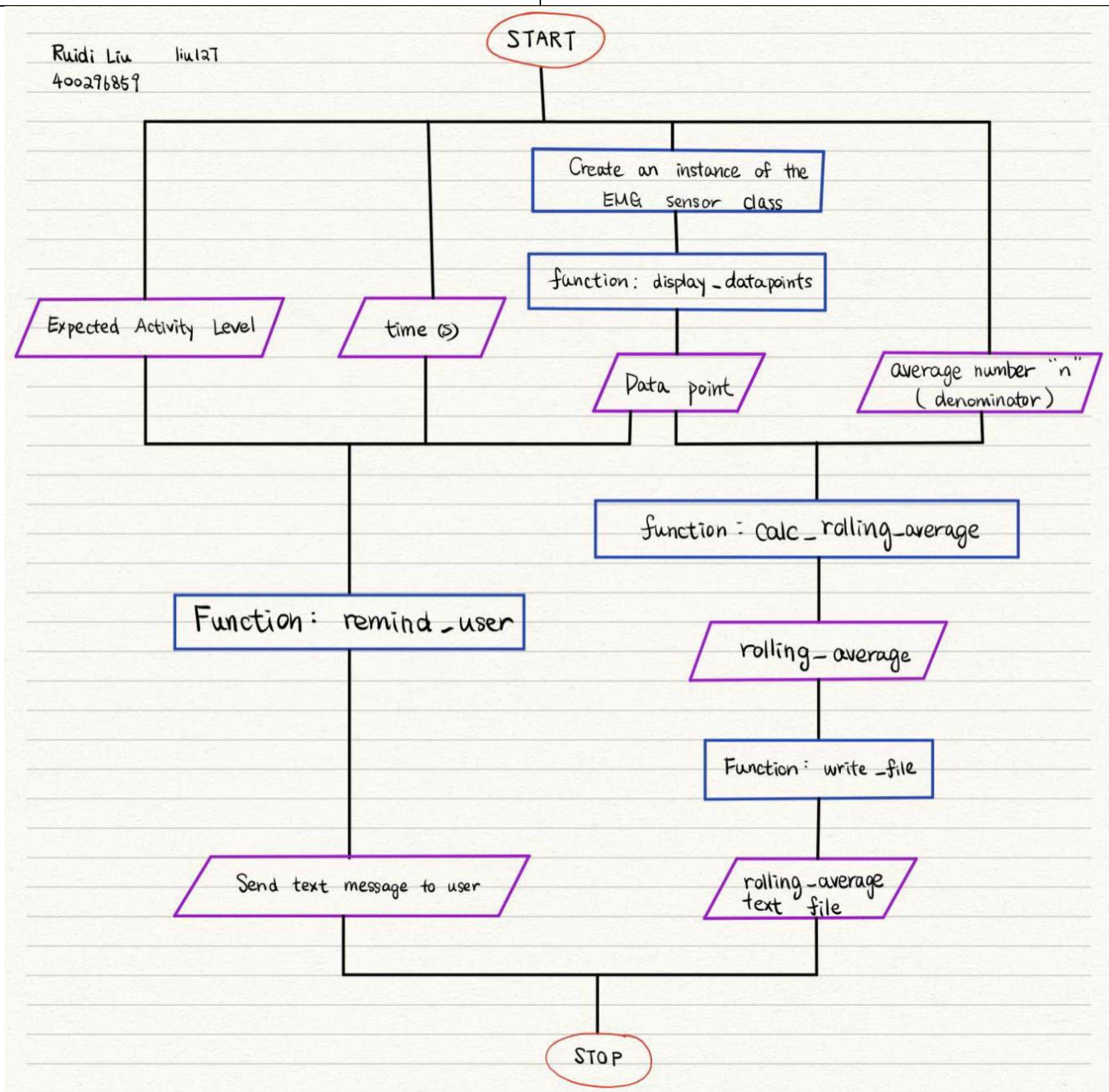
1. Copy-and-paste each team member's screenshots of their flowchart on the following pages (1 sub-team member per page)
   → Be sure to indicate each team member's Name and MacID

We are asking that you submit your work on both worksheets. It does seem redundant, but there are valid reasons for this:

- Each team member needs to submit their flowchart screenshots with the **Milestone Four Individual Worksheets** document so that it can be ***graded***
- Compiling your individual work into this **Milestone Four Team Worksheets** document allows you to readily access your team member's work
    - This will be especially helpful when completing ***Stage 3*** of the milestone

Team Number: 22

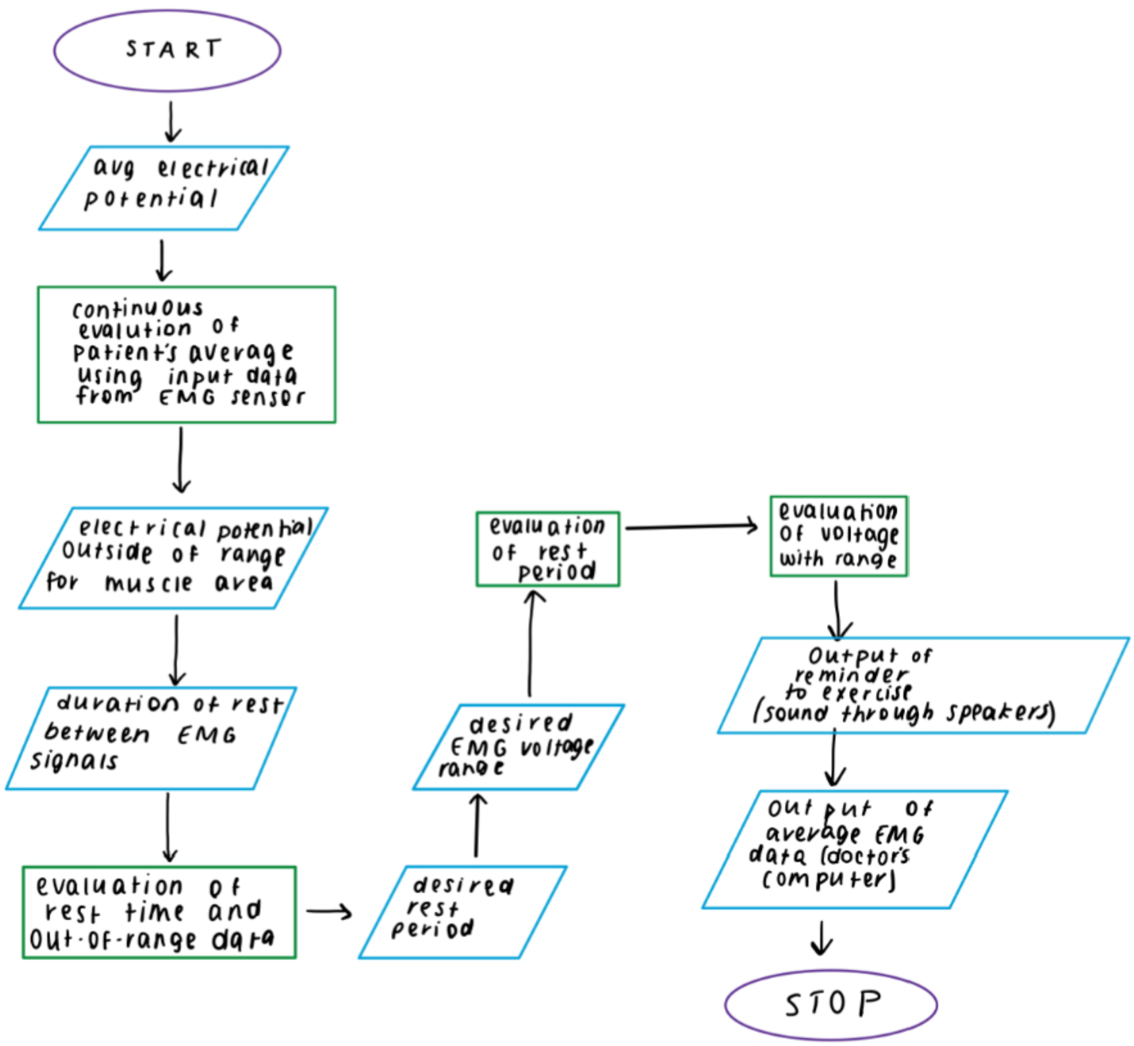
Name: Ruidi Liu	**MacID**:liu127



Team Number: 22

Name: *Sharanya Srirangam*	**MacID**: srirans

Insert screenshot(s) of your flowchart below



*If you are in a sub-team of 3, please copy and paste the above on a new page

MILESTONE 4 (STAGE 2) – DETAILED SKETCHES (MODELLING SUB-TEAM)

Team Number:	22

You should have already completed this task individually *prior* to Design Studio 16.

1. Copy-and-paste each team member's detailed sketch of their housing assembly on the following pages (1 team member per page)
   → Be sure to clearly indicate who each sketch belongs to

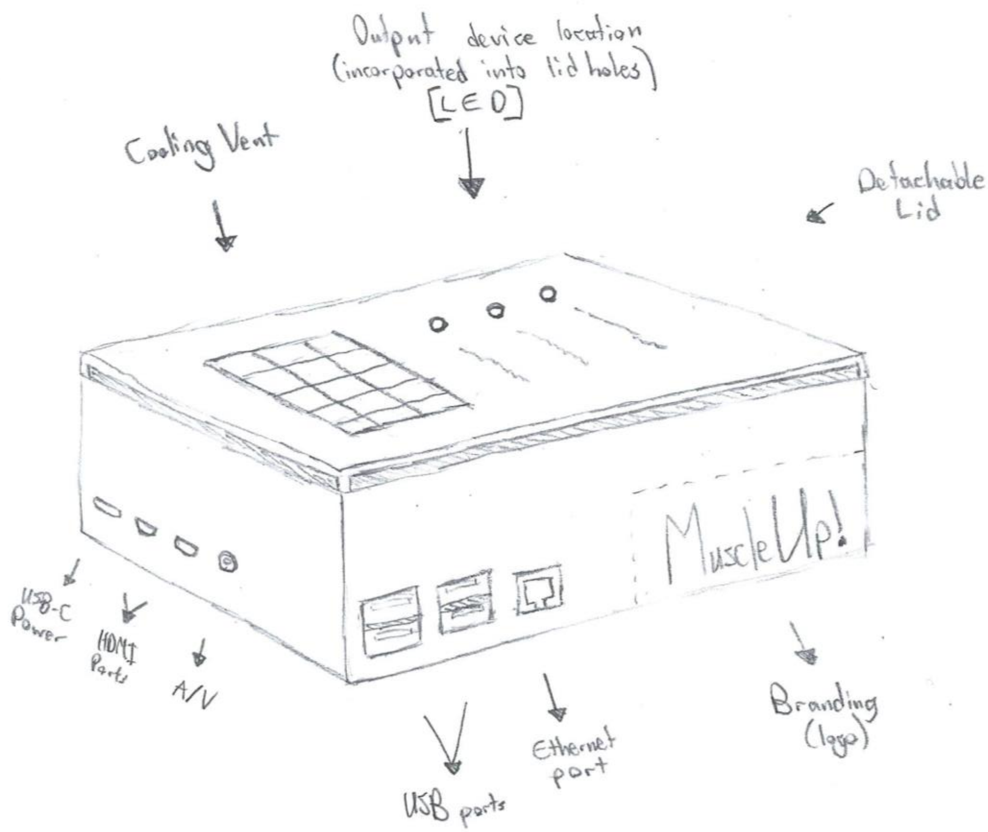
We are asking that you submit your work on both worksheets. It does seem redundant, but there are valid reasons for this:

- Each team member needs to submit their detailed sketch screenshots with the **Milestone Four Individual Worksheets** document so that it can be ***graded***
- Compiling your individual work into this **Milestone Four Team Worksheets** document allows you to readily access your team member's work
  - This will be especially helpful when completing ***Stage 4*** of the milestone

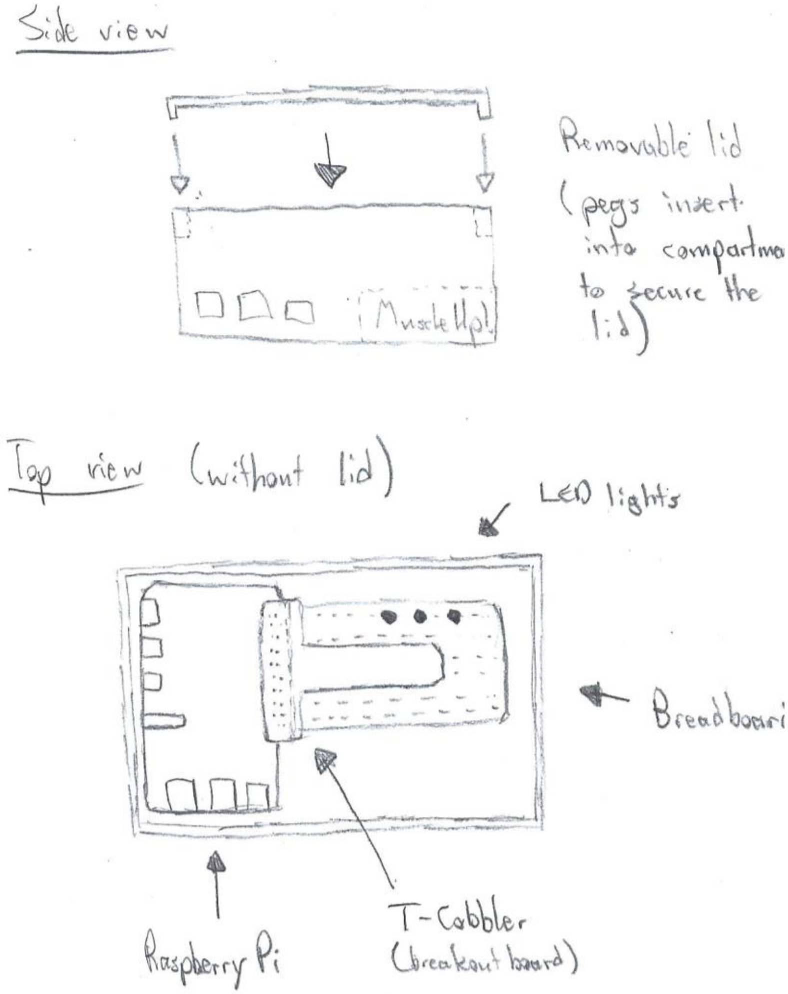
Team Number: 22

Name: *Matthew Ho*	**MacID**: hom23

Pi Housing Assembly



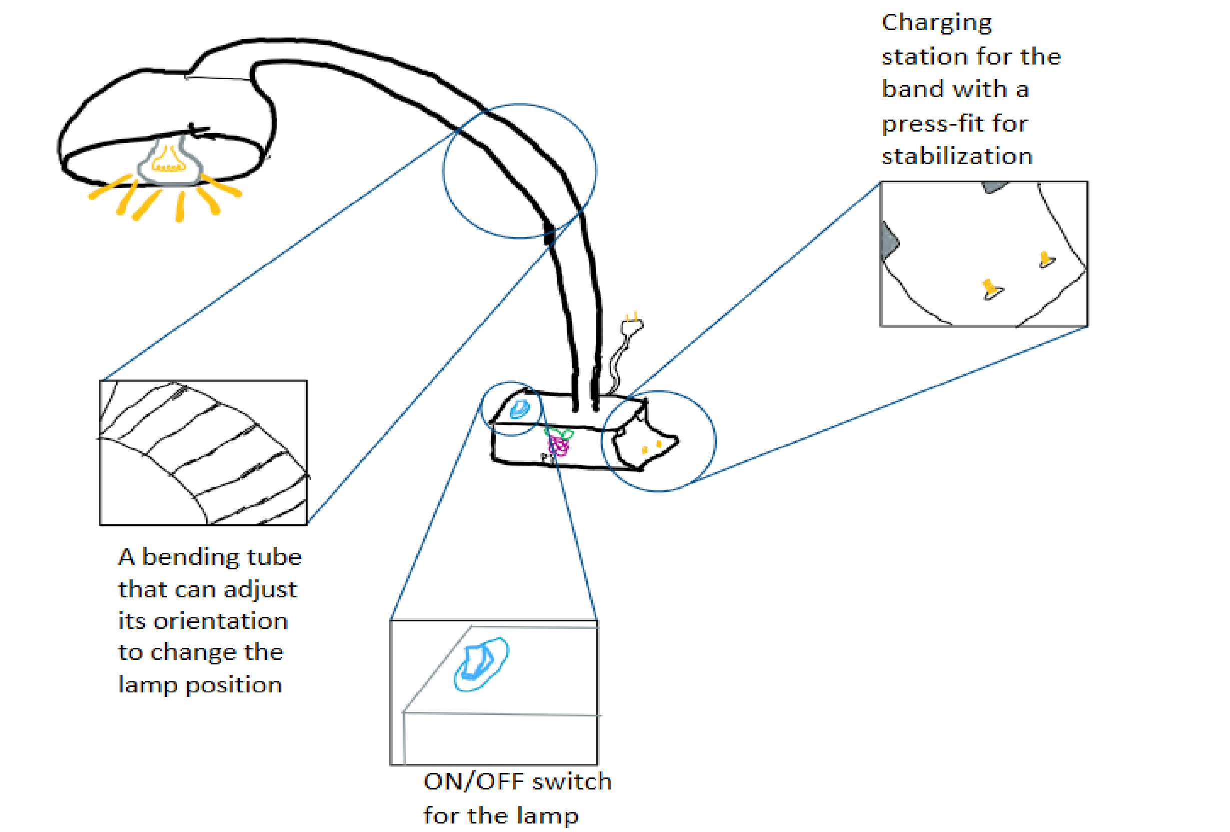
Name: *Matthew Ho*	**MacID**: hom23



*** Words cut off: "pegs inserted into compartment to secure the lid" and "Breadboard"*

Team Number: 22

Name: *Mayar Aljayoush*	**MacID**: Aljayoum



*If you are in a sub-team of 3, please copy and paste the above on a new page

MILESTONE 4 (STAGE 3A) – WORKFLOW PEER-REVIEW (COMPUTATION SUB-TEAM)

Team Number: 22

As a team, document your observations, specifically any similarities and differences between each team member's flowchart in the space below.

Document your observations for each flowchart here.

Similarities:

Inputs: time, expected activity level, measured voltage (data point)

Functions: calculate the rolling average, remind user

Outputs: reminder to user, send text message to user

Differences:

Functions: continuous display of datapoints, writing the rolling average into a text file

Outputs: outputting rolling average list as a text file, sending data to doctor

As a team, write out a pseudocode outlining the *high-level workflow* of your computer program in the space below.

```
Import time
Main()

        Create an instance of the EMG sensor class
        Initialize an empty list named data_list
        Initialize an empty list named rolling_average
        Set value average number = n (could be changed by the user)
        While True:
                Read an instantaneous data point of the scaled data
                Print data_point

                Input data points
                Define function append_datapoints:
                        Append data points to data_list

                If i < length of data list - average number +1:
                        Define function calc_dataset_average:
                                Find the (i+1)th n values in the data set *
```

```
                    Calculate the sum of the values in the (i+1) set
                    Append the calculated average to the rolling_average list
          Else:
               Return rolling_average
          Write rolling_average as a text file
          Return text file

*e.g if average number = 3, find the 4th set of 3 data points in the list

     Define function calc_time_rest
          For every data point of voltage
               If data point ≤ upper_resting_EMG
                    Start time
                    Print time
                    While time_period_rest > prescribed_time_interval
                         Output buzzer activation as a reminder (Buzzer on)
                         If data point > upper_resting_EMG
                              Turn off buzzer
                                   Break
                         Otherwise
                              Wait for 2 minutes before repeating buzzer
                              Wait for 10 minutes before repeating buzzer
                              Wait for 1hr before repeating buzzer (repeating)

     If muscle_activity < expected_lower_bound or muscle_activity >
     expected_upper_bound
          Output message to doctor's phone
```

MILESTONE 4 (STAGE 3B) – PRELIMINARY CODE (COMPUTATION SUB-TEAM)

Team Number: 22

As a team, begin writing your computer program in Python, documenting your progress via screenshots in the space below.

Insert screenshot(s) of your code below

```
import time

sensor = Muscle_Sensor()
i = 0
data_list = []
rolling_average = []
average_number = 5 #This value could be changed

While True:
            data_point = sensor.muscle_scaled(scale)
  print(data_point)

            datalist.append (data_point)

  while i < len(data_set) - average_number + 1:
     data_set = data_list[i : i + average_number]
     set_average = sum(data_set) / average_numebr
     rolling_averages.append(set_average)
     i += 1

print(rolling_averages)

expected_lower_bound = ##enter number here
expected_higher_bound = ##enter number here
upper_resting_EMG = ##enter number here
lower_resting_EMG = ##enter number here
prescribed_time_interval = ##enter number here
buzzer = Buzzer (18)

def calc_time_rest():
   for i in rolling_averages:
   if i <= upper_resting_EMG:
   "Find Code for Start time"

     while time_period_rest > prescribed_time_interval:
      buzzer.on() # add the time for the buzzer
      if i > upper_resting_EMG:
            buzzer.off()
      else:
```

```
 while i < len(data_set) - average_number + 1:
    data_set = data_list[i : i + average_number]
    set_average = sum(data_set) / average_numebr
    rolling_averages.append(set_average)
    i += 1

print(rolling_averages)

expected_lower_bound = ##enter number here
expected_higher_bound = ##enter number here
upper_resting_EMG = ##enter number here
lower_resting_EMG = ##enter number here
prescribed_time_interval = ##enter number here
buzzer = Buzzer (18)

def calc_time_rest():
   for i in rolling_averages:
   if i <= upper_resting_EMG:
   "Find Code for Start time"

     while time_period_rest > prescribed_time_interval:
      buzzer.on() # add the time for the buzzer
      if i > upper_resting_EMG:
            buzzer.off()
      else:
        buzzer.on() # add the time for the buzzer
        buzzer.off()
        time.sleep(120)
        buzzer.on() # add the time for the buzzer
        buzzer.off()
        time.sleep(600)
        buzzer.on() # add the time for the buzzer
        buzzer.off()
        time.sleep(1800)


if data_point < expected_lower_bound or data_point > expected_higher_bound:
   print ("Patient's EMG Voltage is out of range. Please review their data.")
```

* For each screenshot, please copy-and-paste the above to a new page

MILESTONE 4 (STAGE 4A) – HOUSING ASSEMBLY OBSERVATIONS AND EVALUATION (MODELLING SUB-TEAM)

Team Number: 22

As a team, document your observations, specifically any similarities and differences between each team member's detailed sketch in the space below.

Similarities:
- Houses the Pi along with the necessary external ports to connect to the Pi
- Housing placed on the ground (not on the wall or other surfaces)
- Both designs have charging ports for the band

Differences:
- Matthew's includes the breadboard and T-Cobbler (Decided this is unnecessary)
- Mayar's has charging port on the exterior of the box (beneficial for elderly users)
- Mayar's includes a lamp to add texture to the box.
- Matthew's design had a cooling vent which can be a very good solution for pi overheating.

As a team, document the outcome of your design evaluation, including a justification for your decision, in the space below.

- Design should not include the breadboard and T-Cobbler because they are not needed (Buzzer is going to be on body in the EMG band, and text messages to the phone)
- Housing should have the necessary holes to serve as access to the interior, allowing for HDMI, power, USB ports, etc.
- Charging/Holding port for the EMG band will be Mayar's idea of the external port, so elderly patients do not have to struggle with opening the housing lid each time (more convenient this way)
- Housing should stay as a "table mounted" design, not wall mounted
- Model should not include the lamp, but we need a method of making this box more unique than the generic design that already exists
    - Branding on the box (extruded into the box in the CAD model)
    - Possibly a lamp, USB ports to charge phones, etc. that allow for a multifunctioning box
    - Cooling vent (similar to the vents underneath laptops, etc.)
    - Compartments on the top of the housing, users can store items there (funko pops, pens/pencils,keys, etc.)

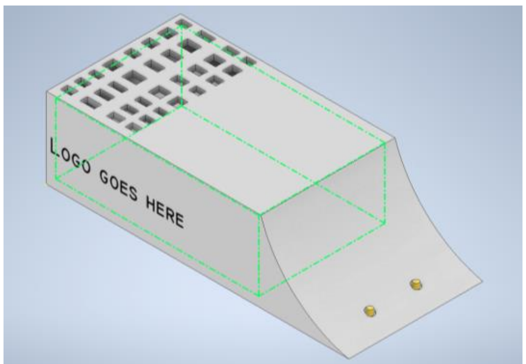
*** Please check notes on the last page of this document for design evalulation conducted with Parmveer!*

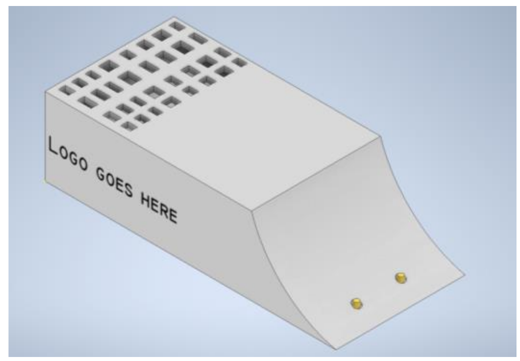
*You do ***not*** need to document the process of your design evaluation!

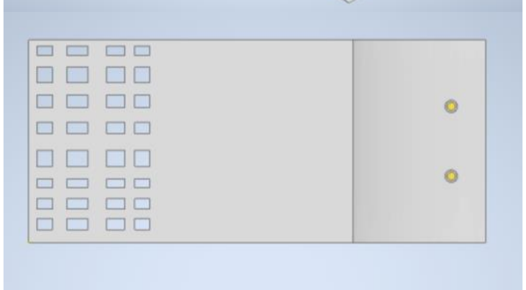
MILESTONE 4 (STAGE 4B) – PRELIMINARY MODELLING (MODELLING SUB-TEAM)

Team Number: 22

As a team, begin modelling your Raspberry Pi Housing components in Autodesk Inventor, documenting your progress via screenshots in the space below.







We will improve the venting when we do the actual model.
We might still add more texture to thee pi housing to have a more unique design.
The design most likely won't include a breadboard and a T-cobbler

* For each screenshot, please copy-and-paste the above to a new page

Notes from Parm regarding wireless connection between EMG Band and Raspberry Pi:

- Wires in the prototype (later on say wireless when Bluetooth can be incorporated)
- ESP 32, Raspberry Pi: RP400 (Bluetooth)
- GPIO Zero (module on Python) to control pins on Raspberry Pi
- Cannot send Python coding over Bluetooth
    - Must have computer on the receiving end (buzzer end) in order for it to work
- We would have a Bluetooth module on the strap that connects to the Raspberry Pi Bluetooth (ideally)
- Bluetooth module with microcontroller AND buzzer in the band AND battery

<u>In real life</u>:

- Raspberry Pi Pico (runs microPython)

Final Verdict:

- Pi send commands wirelessly to the “blackbox” on the EMG band (that is wired to the sensors) that will instruct the EMG sensor and the Buzzer what to do, then data from those 2 sensors will be sent back to the Pi for processing
    - (Blackbox could be the Raspberry Pi pico)
    - Include a little space on the band for this “blackbox” and wire
    - Not a huge space, these Bluetooth boxes are not too large

Make an assembly (Lid + housing)